PI 5 POE NVME CASE

A slide-in tube for a Raspberry Pi 5 with the **Waveshare PoE HAT (G)** on its header and a **Pimoroni NVMe Base Duo** underneath. No screws or standoffs in either board: **each board rests on its own ledge**, the PCIe ribbon is the only link between them, and a **click-in PLA lid** closes the SD end: its **tray carries the Duo from below** and two corner posts stop both boards against the closed USB end. The body prints standing on its USB end, so the ledges, vents and ports need no supports.

PoE HAT (G) · WS-27670 | NVMe Base Duo · PIM704 | Pi 5 + Active Cooler | PETG or PLA body · PLA lid · TPU feet | Prusa MK3S+

OUTSIDE L × W × H	HEIGHT ON FEET	PARTS	PRINT, ALL PARTS
106.3 × 68.8 × 64.6	73	body · lid · 4 feet or DIN bracket	6h 58m

Changes, 13 Nov

- **The SSDs sit on top of the Duo**, under the Pi (your correction). The bay under the Duo that was kept for them is free.
- **Lid v0.5, PLA: a tray under the Duo.** It's slotted, 2 mm thick, and reaches 63.8 mm under the Duo (75 %). Webs run underneath to just above the floor. Two corner posts put a small lip over the Duo's top corners and stop both boards. It replaces the TPU card retainer, and the two can't be fitted together.
- **Feet v0.5: M4 × 10 button heads straight into the PLA.** The 4 mm floor holes were drawn for M3 heat-set inserts, and an M4 cuts its own thread in them. The feet are 3 mm taller, at 9 mm, with a Ø7.6 × 3.8 counterbore. The screw stops 0.6 mm short of the end of the hole.
- **Next body:** a much wider Duo ledge (your note). It can also lose the empty bay under the Duo.
- **To print: the lid and the feet.** The body stays as printed. Earlier: on 11 Nov the body printed well and the ledges held the Duo through a shake test. The TPU lid's snap bumps didn't print, which led to the PLA latch lid (v0.3).

Settled

- **SSDs: Samsung 980 series, on top of the Duo.** Single-sided, 2.38 mm, no heatsink. On Pimoroni's 5 mm SSD standoffs they stand about 7.4 mm above the Duo, in the 12 mm gap under the Pi.
- **HAT height: unknown, so kept at 16 mm.** That's the tall case. If your HAT sits lower, you just get more air above it.
- **HAT stays on its headers.** No hold-down ribs.
- **SD tray dropped.** Fit the microSD card before the PCIe ribbon.

The print-ready STLs are attached to this PDF (open the attachments or paperclip panel in your PDF reader). Each is already oriented for the bed:

- `pi5_case_body_PETG.stl · 374 KB`
- `pi5_case_lid_PLA.stl · 126 KB`
- `pi5_case_feet_x4_TPU.stl · 94 KB`
- `pi5_case_din_bracket_PETG.stl · 133 KB`

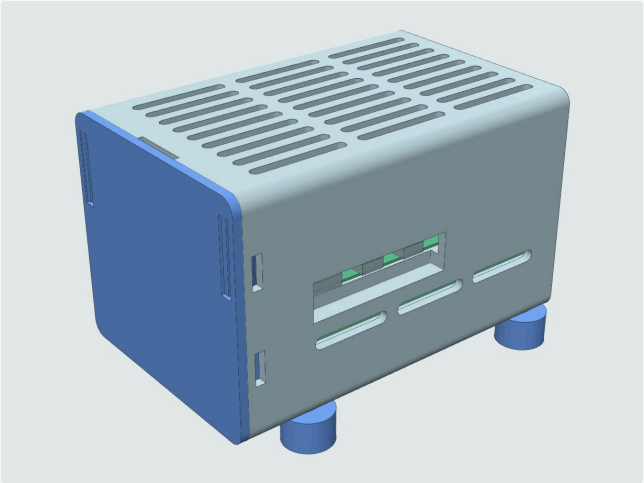
Print time

PRUSASLICER 2.7 · 0.30 MM DRAFT (LID 0.20 MM TO THE HOOKS, THEN 0.30) · 0.4 NOZZLE · MK3S+

	Material	Time
Body	PETG, USB end on the bed	**4h 26m** 77 g
Lid	PLA, outer face on the bed, tray standing up (G-code in `gcode/`)	**2h 03m** 26 g
Feet × 4	TPU, eSun 95A, slower preset, one at a time (G-code in `gcode/`)	**28m** 4 g
DIN bracket	PETG, rail side down. Draft: 1h 39m. Recommended 4 walls, 40% gyroid	**2h 24m** 43 g
Case + feet		**6h 58m**

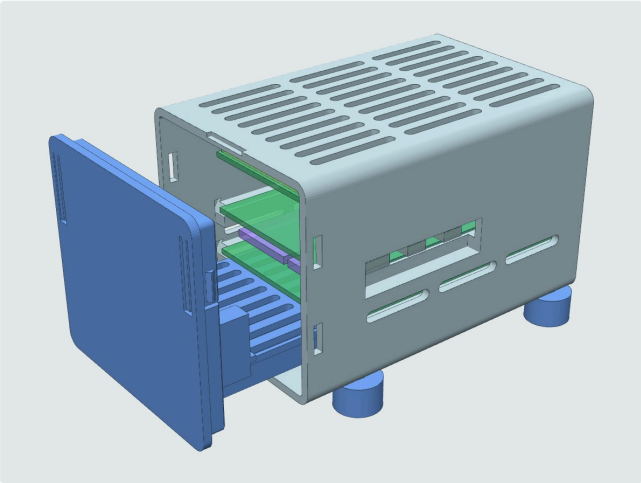
Walls, roof, floor and end are 2.4 mm. The same body with your usual 3 mm walls slices at 5h 23m, so 2.4 saves 56m. The stiffness comes from the two lips and the thick SSD-bay walls, which run the full length on both sides like rails, and from the bars between the vent rows.

3D views



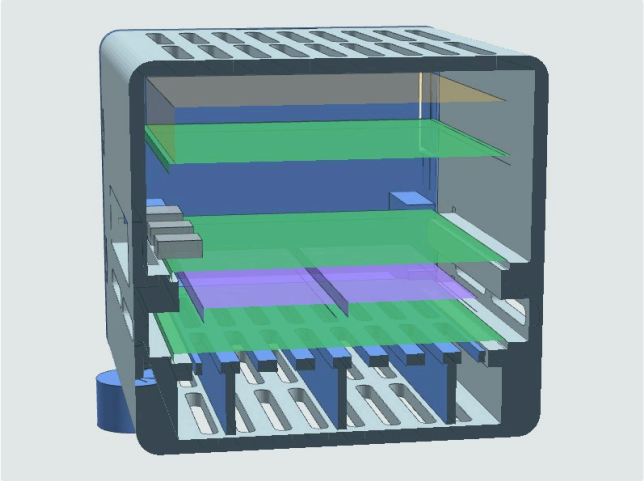
ASSEMBLY

Case on its feet with the boards in place (boards, SSDs and HAT parts shown as blocks).



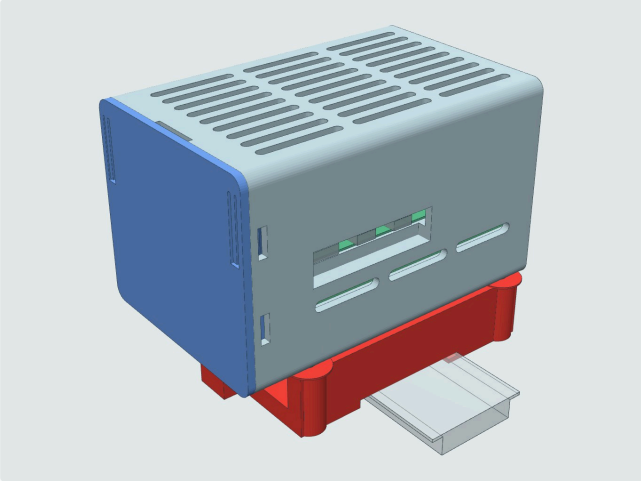
LID OUT

The PLA lid pulled 30 mm off the SD end, its tray sliding out from under the Duo.



SECTION THROUGH THE LEDGES

Cut across the case: the Pi on the upper ledge, the NVMe Base Duo on the lower ledge with the SSDs on top, and the lid's tray under it.



ON A DIN RAIL

Feet replaced by the DIN rail bracket, clipped to a TS35 rail.

Rendered from the 3D model. The interactive viewer is a separate page.

Drawings

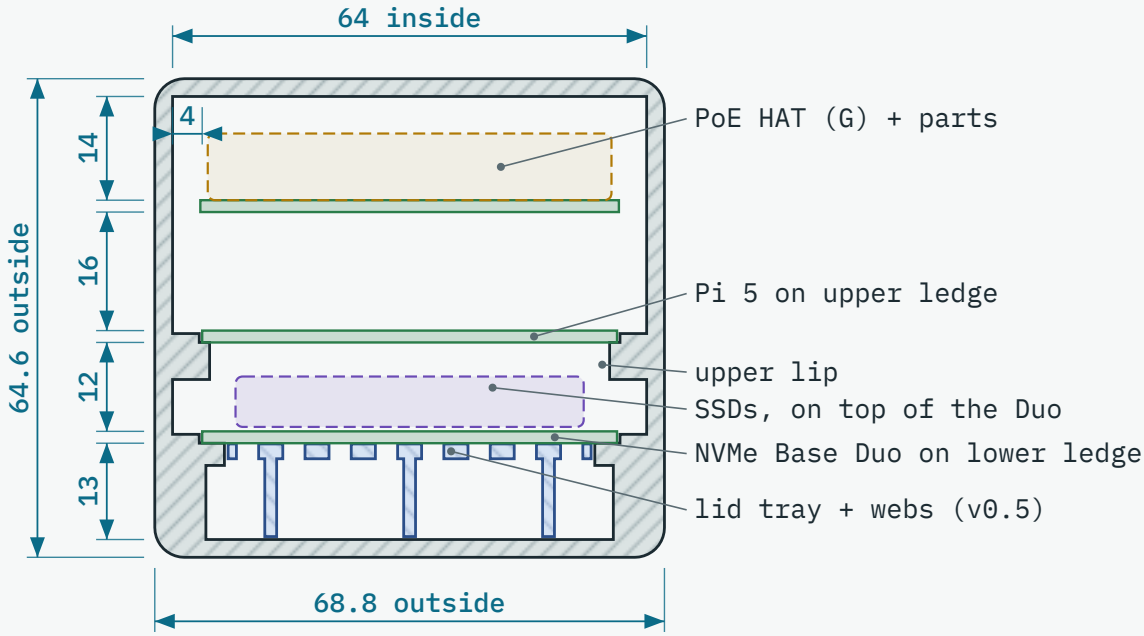
Dimensions in mm. The drawings are not to scale on the page; a scale in a title (3:1, 2:1) is relative to the other views.

body (cut) lid / TPU parts PCB SSD zone HAT parts PCIe ribbon dimension (mm)

CROSS-SECTION

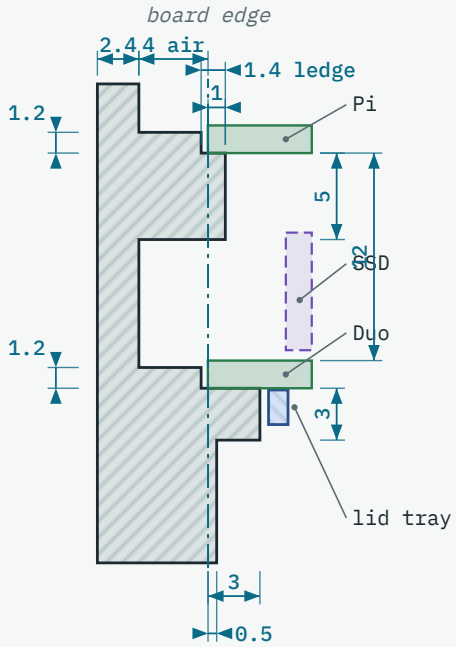
SECTION A–A · ACROSS

Your second sketch. Each board sits on its own ledge and nothing screws into either board. The upper lip runs between the two boards: the Pi rests on its top, and the Duo's edge runs in the recess under it, resting on the lower lip. The SSDs sit on top of the Duo, under the Pi. Below the lower lip the wall steps back out; that space was meant for the SSDs and now takes the lid's tray and webs.



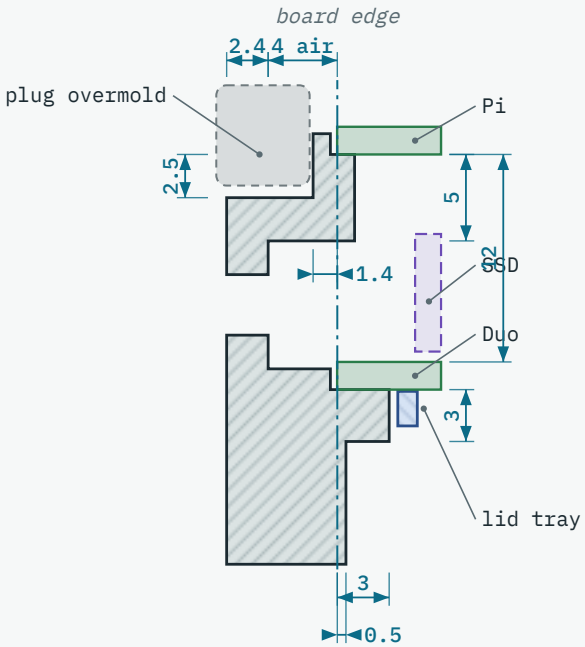
DETAIL · THE LEDGES (3:1)

A 1.2 mm fence on each ledge, lower than the PCB, keeps each board centred with 0.4 mm play a side. The Pi's ledge only reaches 1 mm under it: the GPIO solder joints and the jack tabs on the Pi's underside start about 1.4 mm in from the edge. The Duo's ledge reaches 3 mm. That's too narrow (your 13 Nov note), so the lid's tray now carries the Duo from below, and the next body gets a much wider ledge.



DETAIL · LEDGES AT THE PORT WINDOW (3:1)

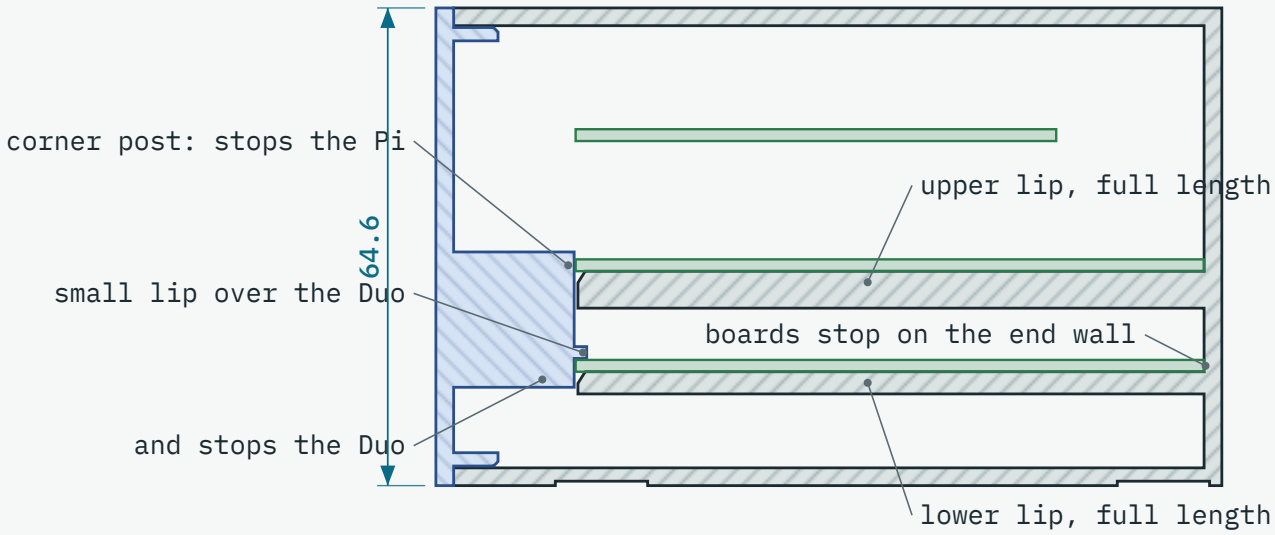
A chunky USB-C plug's overmold hangs about 2 mm below the Pi's board edge. Along the 42 mm port window the upper lip's outer half drops 2.5 mm so the plug can seat; the 1 mm ledge under the Pi carries on, tied to the wall lower down.



LENGTHWISE

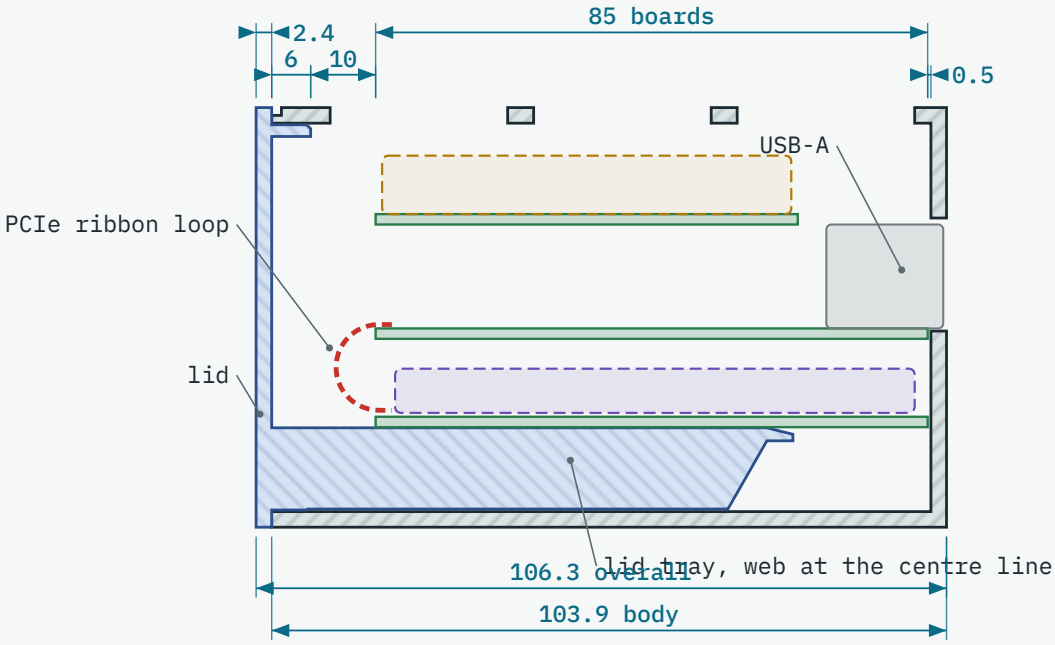
SECTION C–C · ALONG THE GPIO-SIDE LEDGES

Both lips run the full length. The boards slide in from the SD end, USB first, together (the ribbon joins them) and stop on the closed end wall. The lid's corner post (v0.5) stops both boards at the SD end, and its small lip sits over the Duo's top corner.



SECTION B–B · DOWN THE MIDDLE

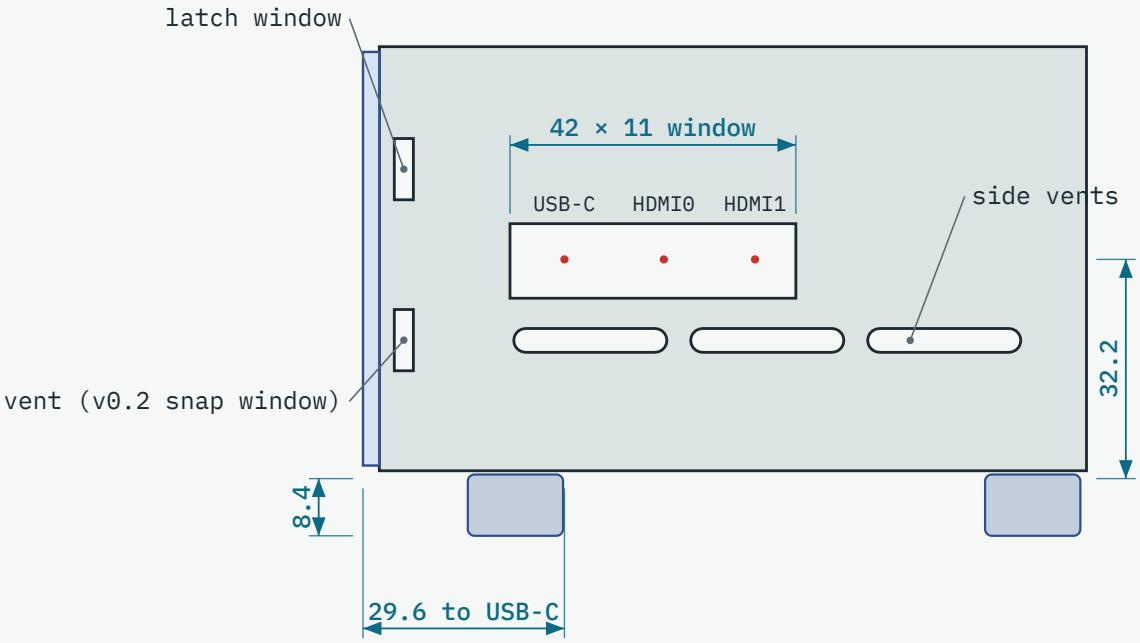
Centre line, through the first USB stack. The PCIe ribbon loops round the SD end of the boards; there is 10 mm for it between the boards and the lid. Under the Duo, the lid's tray, cut here along its middle web.



OUTSIDE

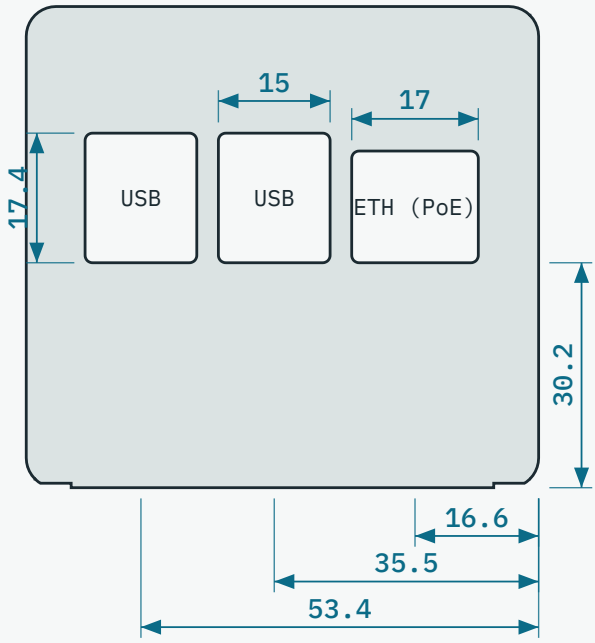
SIDE ELEVATION · HDMI / USB-C SIDE

One 42 × 11 window covers USB-C and both micro HDMI, sized for a dock cable's overmold. Below it, slots vent the space between the two boards. Near the lid, the upper small window takes the lid's latch hook; the lower one is now just a vent. The other long side has the same vents without the port window.



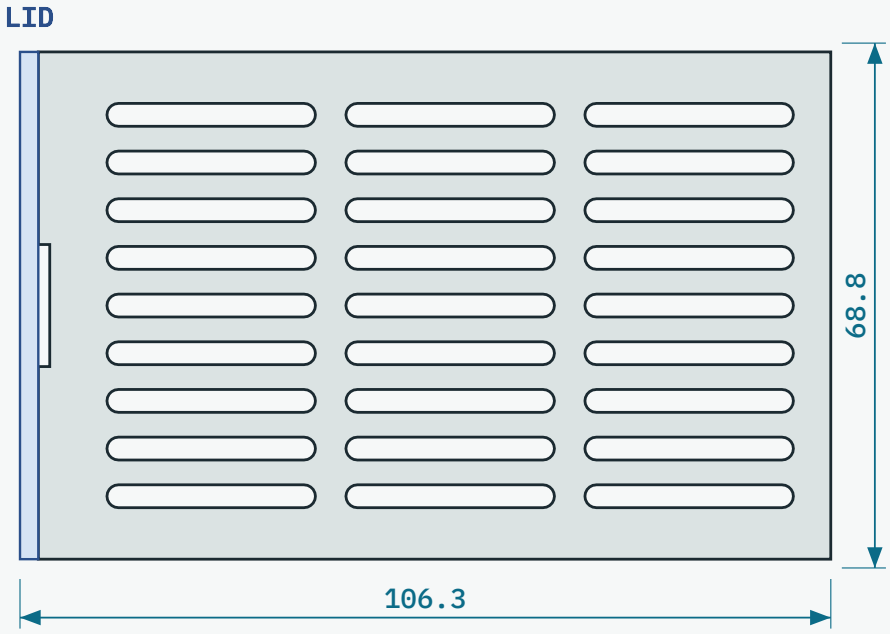
USB / RJ45 END

The closed end of the body. It prints first, flat on the bed. Cut-outs are 0.5 mm clear of each jack.



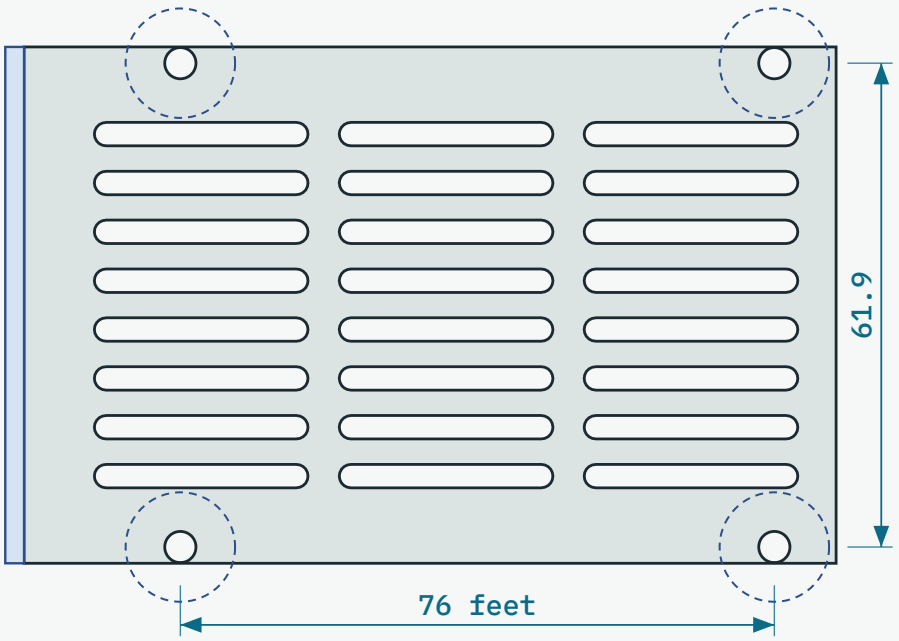
TOP

Roof vents: 3 mm slots on a 6.25 mm pitch in three rows. The bars between rows stiffen the roof.



BOTTOM

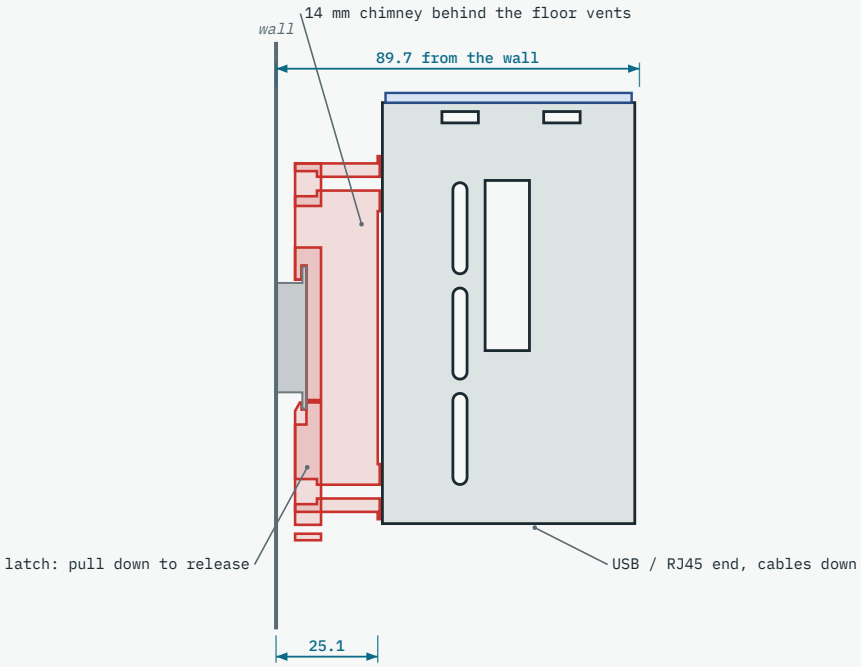
Floor vents under the Duo (the lid's tray is slotted over them). The four feet screw into 4 mm holes in the thick lower walls, inside 0.6 mm locating recesses: M4 × 10 screws cut their own thread in the PLA.



DIN RAIL BRACKET

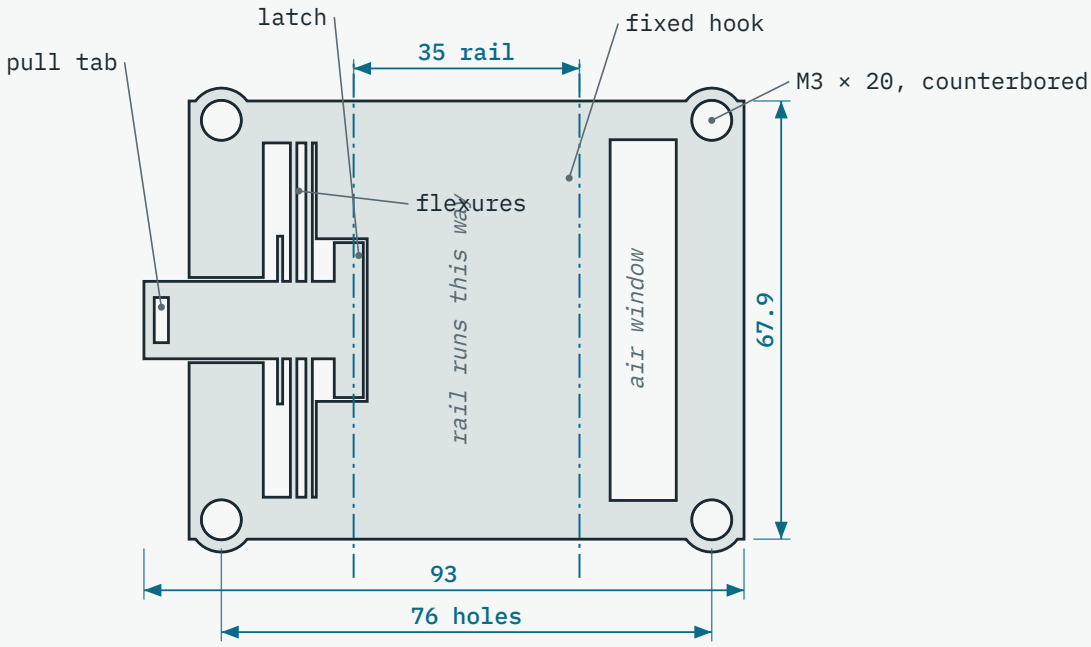
ON THE RAIL · SIDE VIEW

Optional mount in place of the feet. The case stands with its length vertical, here USB end down so cables hang (the bolt pattern is symmetric, so it also bolts on the other way up). Its floor vents open into a 14 mm chimney behind it.



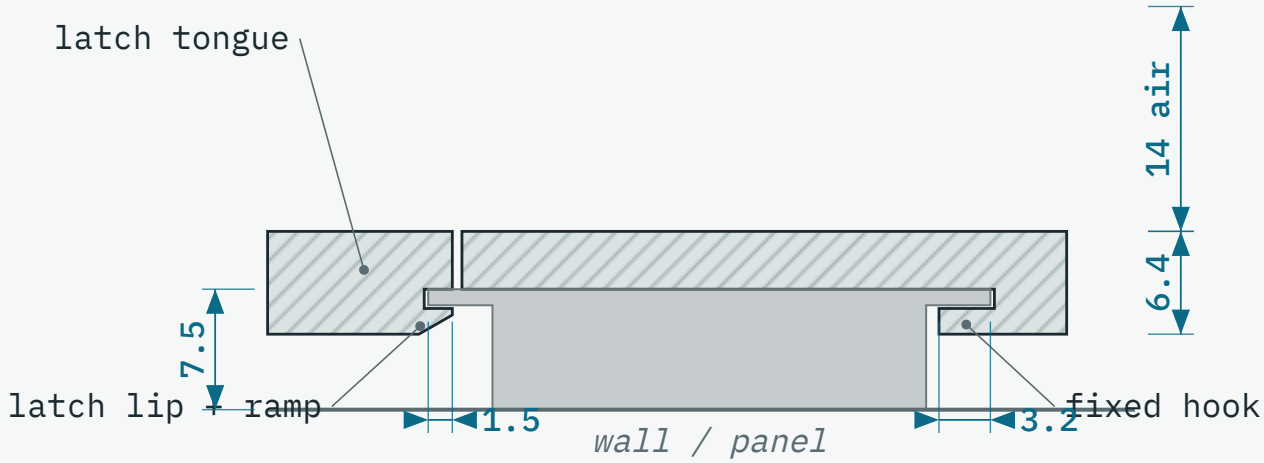
BRACKET · PLAN

Seen from the case side, cut through the plate. Four M3 × 20 socket heads go through counterbored bosses into the foot inserts. The tall side ribs run under the case's thick lower walls, clear of the floor vents, and stiffen the whole bracket. The window and the flexure gaps let air through.



BRACKET · CLIP ON THE RAIL (2:1)

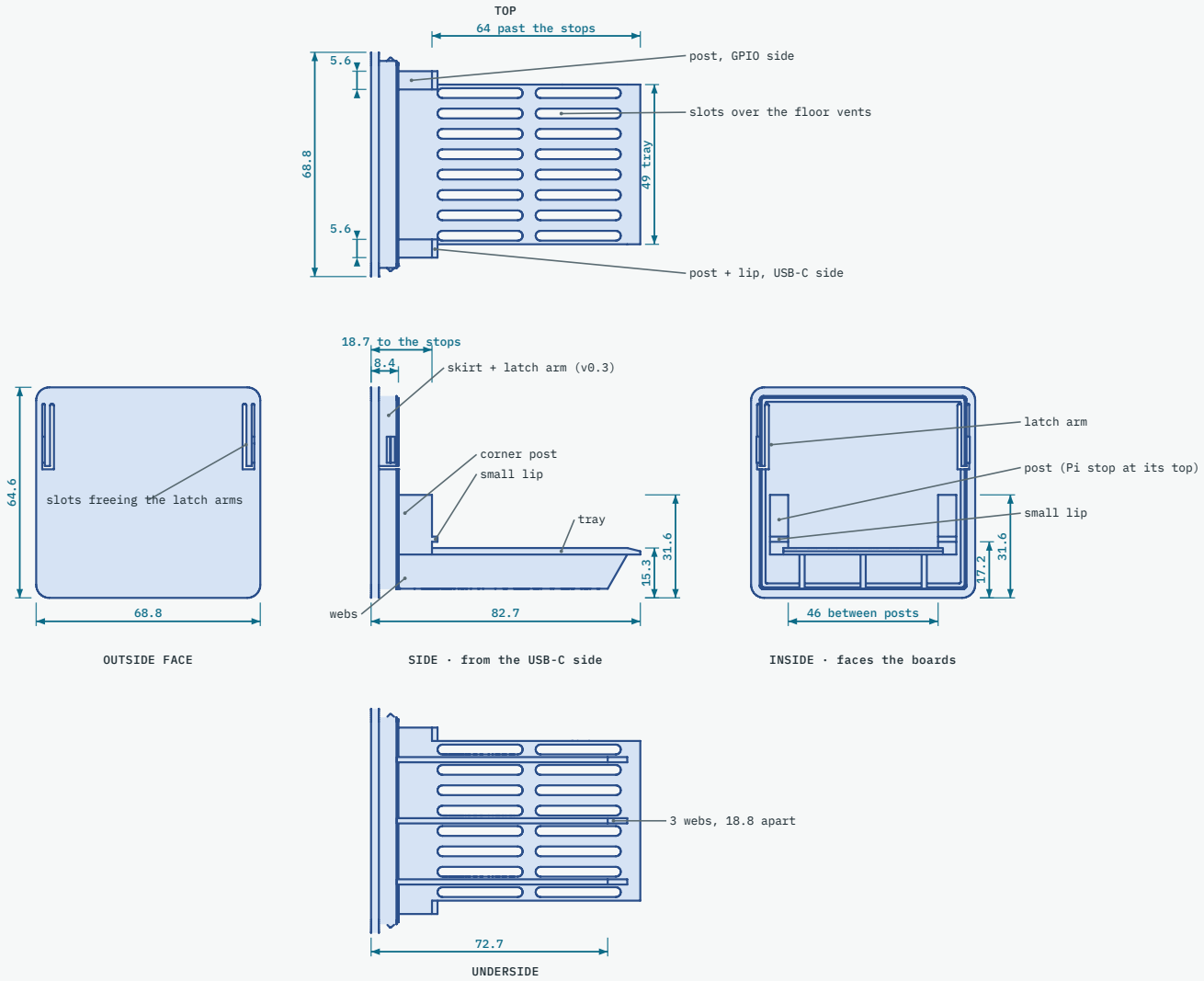
A fixed hook runs the full width on one flange. The other flange gets a latch on four flexures with a lead-in ramp, so it snaps on when pushed. To remove it, lever the pull tab with a screwdriver; a stop limits the latch to 2.2 mm of travel so it can't be overbent, and the hook stays engaged even then. Rail contact is spread across the bracket's full width.



LID V0.5 · TRAY UNDER THE DUO

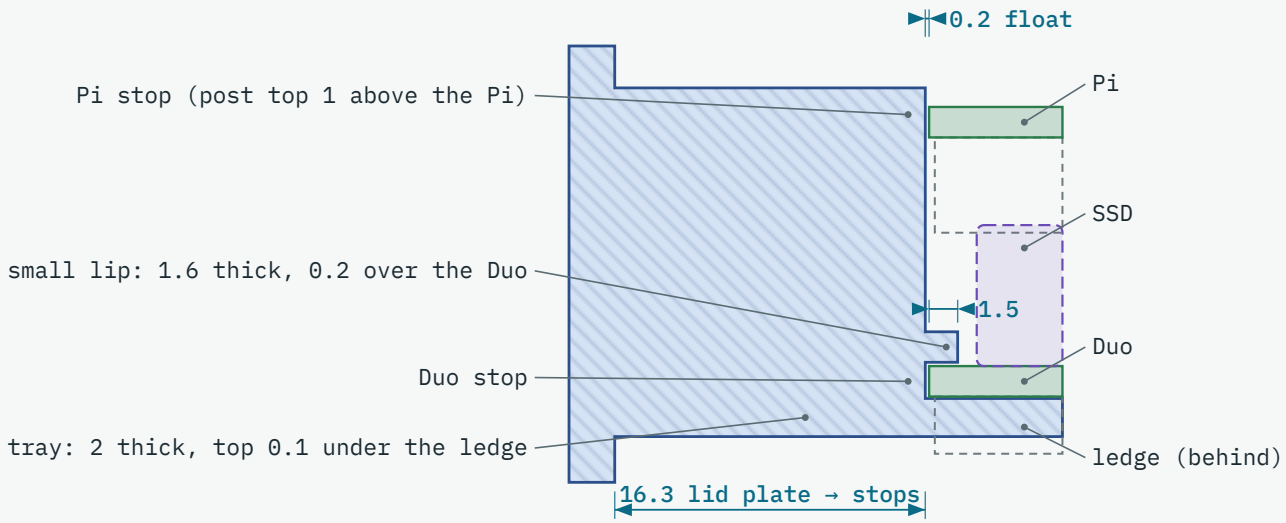
LID · ORTHOGRAPHIC VIEWS

Third-angle projection, all at one scale. The side view is in the middle, with the top above, the underside below, the face that goes inside the case on the right and the outer face on the left. Heights are measured from the lid's bottom edge, lengths from its outer face.



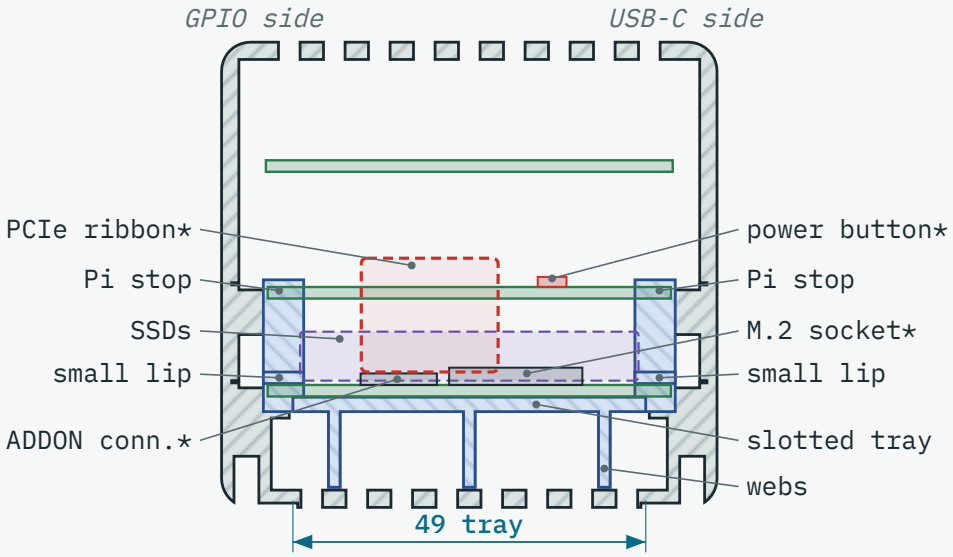
CORNER POST · SECTION (3:1)

One post in each corner does three jobs. The tray holds the Duo from below. The small lip sits 0.2 mm over the Duo's top corner and reaches 1.5 mm in. The post's face stops both the Duo and the Pi, with 0.2 mm float so the latches still click. In the other views the post is only 5.6 mm wide.



FROM THE LID END · LIKE YOUR PHOTO

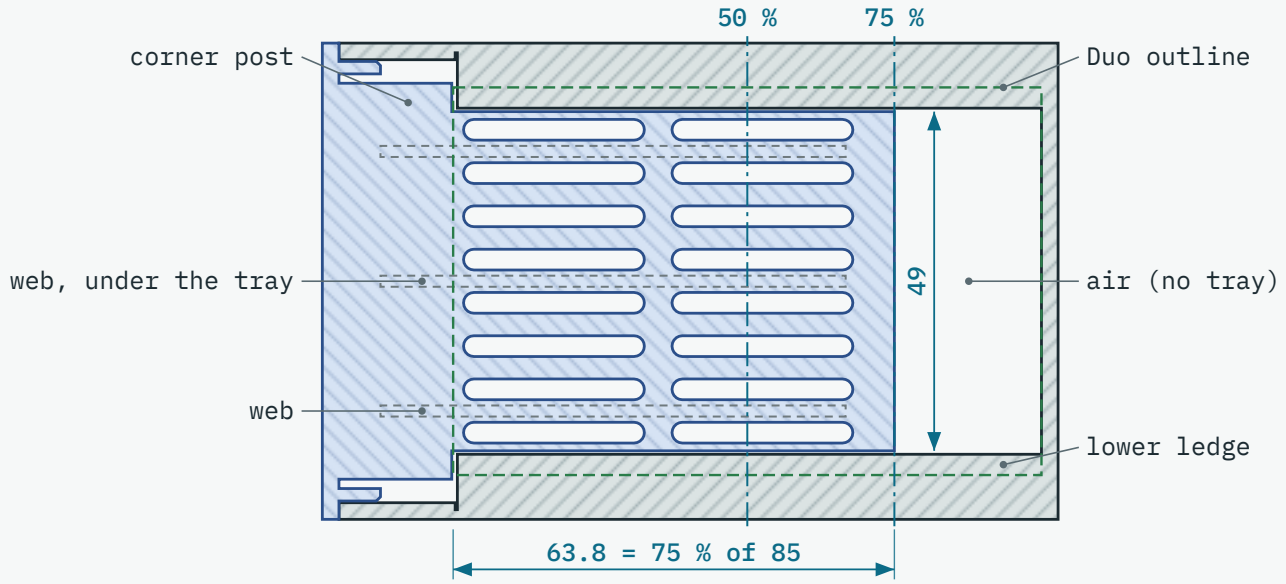
The posts sit in the two corners you circled, clear of the PCIe ribbon, the M.2 socket, the ADDON connector and the Pi's power button (positions read off your photo).



** approx., from your photo*

PLAN UNDER THE DUO

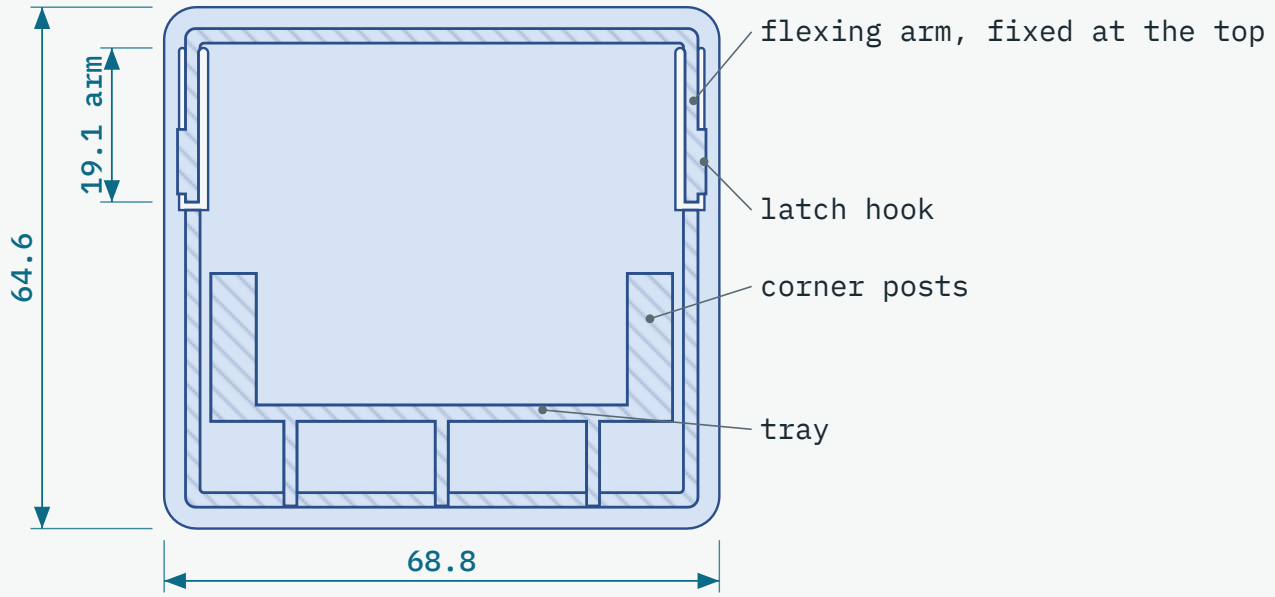
The tray runs between the lower ledges, 0.5 mm clear each side, and reaches 63.8 mm under the Duo (75 %). Its slots line up with the floor vents. Three webs underneath run on the floor's bars and stop 0.4 mm above the floor, so the tip can't droop.



LID AND FEET

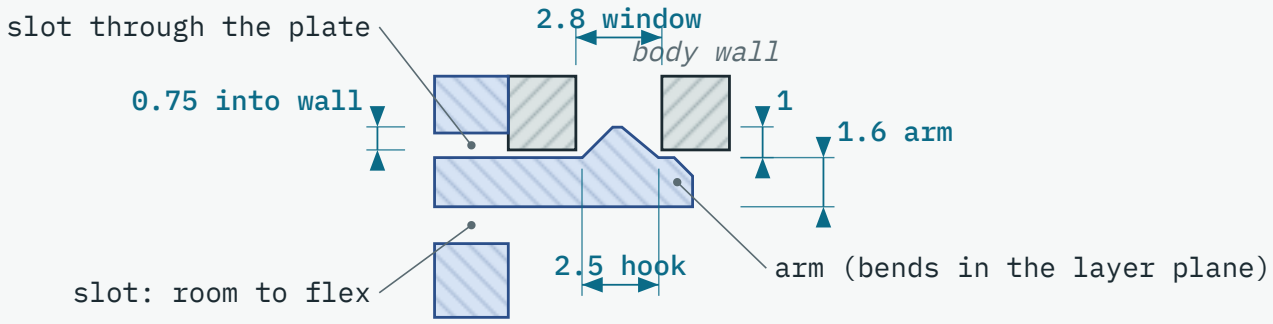
LID · INSIDE FACE

v0.5, PLA. A 6 mm skirt locates it in the mouth. On each long side, slots cut a 19 mm arm out of the skirt, fixed at its top end; its hook clicks into the body's upper side window. Cut further in: the two corner posts and the tray with its webs.



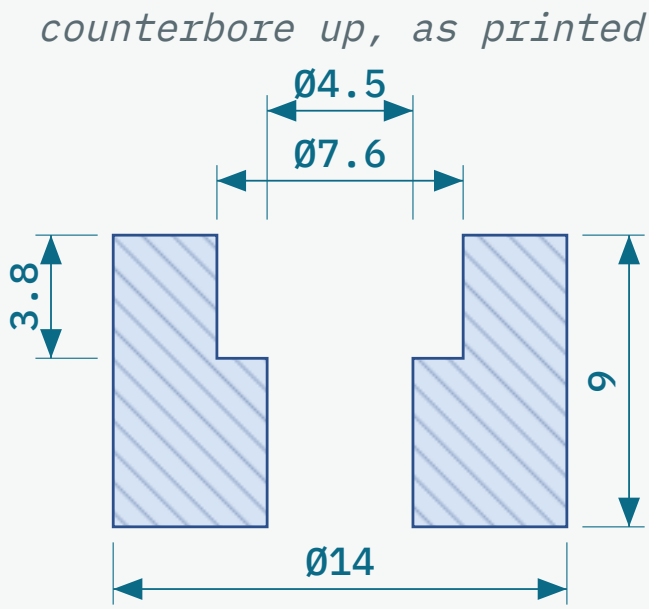
DETAIL · LATCH IN THE WINDOW (4:1)

Plan cut through the hook. The arm is 1.6 mm thick and bends sideways, in the plane of the print layers, so PLA flexes along its filaments; at the 0.75 mm it moves to pass the wall, the strain is about 0.9%. The hook's lower face is a 45° overhang as printed; the lead-in is gentler, so it pushes in more easily than it pulls out.



FOOT

v0.5. TPU, Ø14 × 9, printed counterbore-up. Fit it counterbore toward the desk. An M4 × 10 button head (Ø7) sits in the Ø7.6 × 3.8 counterbore, with 5.2 mm of TPU under it. It cuts 4.8 mm of thread into the body's 4 mm hole and stops 0.6 mm short of the hole's end.



Design notes

WHERE THE SPACE GOES

VALUE	REASON
13 under the Duo	Kept for SSDs on the Duo's underside. They turned out to be on top, so the lid's tray and webs use this space now. A future body can drop it.
12 gap	Duo top to Pi underside. The case sets it now that there are no standoffs; 12 mm is what Pimoroni's ribbon is made for. The upper lip (5 mm) sits in it with 7 mm clear above the Duo. It only reaches 1 mm in, so the SSDs (about 7.4 mm tall) stand clear of it.
1 / 3 ledges	How far each ledge reaches under its board. The Pi's is short to stay clear of the solder joints on its underside.
16 header	Pi top to HAT underside, the usual height that clears the Active Cooler.
9 + 5	Tallest part on the HAT, then air under the roof vents.
4 sides	Air between the boards and the side walls, above the upper lip and in the recess between the lips, as you asked.
10 SD end	Room for the ribbon loop (about 7 mm) between the boards and the lid.
42 × 11	USB-C + HDMI window. A dock cable's USB-C overmold has about 8 mm of window either side and has to reach ~5 mm in to seat the plug.

ASSEMBLY

1. Fit the SSDs to the top of the Duo and the ribbon to both boards as Pimoroni describe, but leave the board standoffs out. Plug the HAT onto the Pi's GPIO and PoE headers. Fit your microSD card now if you use one.
2. Fold the Duo under the Pi and slide both in from the open end, USB first: the Duo onto the lower ledges, the Pi onto the upper ledges. Push them together by their SD edges until they stop on the end wall.
3. Push the lid in. The tray slides under the Duo (its tip lifts the board if it sags), the posts' small lips pass over the Duo's top corners, and both latches click into the upper side windows. The posts stop both boards.
4. Screw the TPU feet on, counterbore toward the desk, with M4 × 10 button heads straight into the floor holes. They cut their own thread, so go gently the first time, and stop when the head is snug.
5. To open, pull the lid out by the thumb notch at the top. If it's tight, press both hooks in through the side windows with a small screwdriver as you pull. The boards then slide out by their SD edges.

PRINTING

- Body: USB end flat on the bed, PETG. Every wall, the lip and the drive-bay walls are vertical in that orientation. Vent slots and windows only bridge 3–10 mm. No supports, no brim.
- Lid: outer face on the bed, PLA, no supports, no brim. Layers are 0.20 mm up to 9 mm, where the latch hooks are, then 0.30 mm. The tray, webs and posts stand straight up, about 83 mm tall, and each layer stays inside the one below. The arm slots go right through the plate; check they're clear on the first layer. The hooks' undersides are 45° overhangs.
- Feet: TPU, counterbore side up (no overhang), printed one at a time. Turn them over to fit.
- The click is set by `latch_t` (1.6, arm thickness) and `latch_hook` (1). Too stiff: thin the arm to 1.2. Too loose: raise the hook to 1.2. Either way, reprint only the lid (2h 03m).

HARDWARE

- 4 × M4 × 10 button head screws (feet), straight into the PLA. No inserts.
- DIN mount instead of feet (not yet changed for M4): 4 × M3 × 20 button or socket head screws into M3 heat-set inserts, TS35 rail (35 × 7.5 or 35 × 15)
- Nothing for the boards. The Duo kit's standoffs and screws aren't used.

YOUR BRIEF (VERBATIM)

> Let’s make a raspberry pi 5 case. Requirements:
> - it has a PoE hat (check my Gmail inbox for order confirmations from Core Electronics for the part number)
> - it has a pimoroni dual nvme base
>
> I want to avoid screw holes, I would prefer. Refer to this sketch of the side profile showing how I would like the pi and the nvme base to sit on a lip
>
> The nvme base will need enough vertical clearance from the pi to account for thick nvme disks (dual Samsung 1TB)
>
> For the USBC port, it will need enough clearance to account for fairly chunky usb connectors like the ones on dell thunderbolt docks
>
> Previous designs I have made use. 3mm walls but that’s not a hard requirement. If strength can be gained through smart use of webbing and structural ribs then that would be ideal if print time can be optimised
>
> Printing in 2-3 parts would be ideal,
> - main body
> - lid
>
> If the lid is on the smallest face (the usb/rj45 end) or the sd card end that would be ideal. The pi can slide in like into the case as if it were a tube
>
> Attaching the end pieces “lids” can wither be screw in (m3, counter bore heads, or a nice click in place) for a screwless click in place I would print that from TPU to allow for some flex with whatever you use, and it can still retain some rigidity
>
> Give it ventilation holes top and bottom, you can increase the space between the inner walls and the nvme base/poe hat as they tend to generate a bit of heat
>
> It will also need some feet that can be screwed in with m3 counterbored screws
>
> Make some sort of tray for the as card so it can be slotted in and out like a SIM card
>
> I think those are all the requirements for now
>
> Create 2d projections before the 3d model

With a first sketch of the side profile: thin upper wall, a lip the board sits on, a thicker wall below.

Follow-ups:

> I'd prefer the cross section be like this [second sketch]
> I don't want to screw into the pi or the nvme base, the idea is for this to be simple
> Alright skip the sad card holder stuff
> 1. Samsung 980EVO
> 2. No idea
> 3. Not fussed, the gpio pins hold it in pretty snugly
> Let's create another component that uses the footprint of the feet to attach it to the case.
> It will be a bracket to mount it on a DIN rail
> Leave plenty of clearance for the airflow, make sure the bracket is sturdy and rigid as people may be rough installing and removing it
>
> 11 Nov, after printing:
> The case printed well, the little tabs on the lid did not print at all
> PLA might be able to do the job if you make the tabs thin enough to flex a little
> The rails are doing a good job of securing the nvme base
> I can shake the case and the base is secure
> Can you confirm your design of the feet suit these screws [2 photos: button-head screw in calipers]
> Then I think securing the cards should be separate from the back lid. So we make a TPU insert that locks the cards in tightly, then the pla lid goes on after that
> If there's a better way to do that then by all means suggest it and don't be 100% stuck to the shape of this bit [photo: sketch of a frame in the mouth pressing the board edges]
>
> 13 Nov:
> Got some lid changes
> - The drives face up, and sit between the pi and base. Your design assumes the other way. Which isn't a problem, we can resolve that with a modified lid
> - the ledge for the nvme is too narrow. In future designs it needs to be much wider
> - For now let's create a pla lid that has a tray to support the base for at least 50-75% the length of the card
> - Have a much smaller lip to secure the top of the card
> - another lip to keep the pi pressed in, this lip will press up directly against the pi
>
> Put together some sketches to verify that you understand my intention [photo: the boards in the body, marked up; sketch of the lid from the side]
> Add this to your work list and get to it once we've settled on the

> lid design:
> I’m not sure if you realised it based on my design specs, the screw holes for the TPU feet fit an m4 screw, not an m3 as initially requested.
> The only m4’s I have are a button cap screw with a 7mm head. So the counterbore hole for the feet needs to suit that, and be height of the feet needs to increase 3mm
> Can you explain corner post 3.2? There’s a very large blue shaded area
> If you can do some orthogonal projections of the new lid design that might help
> 1. Yes
> 2. Yes
> 3. Not sure
> 4. Whatever will print more reliably
> [answers: M4 × 10, straight into the PLA]

The second sketch: an upper lip with the Pi resting on top, a recess in the wall where the NVMe board's edge runs, a lower lip the NVMe board rests on, then the wall steps back out below.

Source: `pi5_case.scad` (OpenSCAD, every value above is a parameter) and a handoff README, saved in the Designing stuff project under `pi5-poe-nvme-case/`. Drawings are cut from the model's STLs.

Part data: PoE HAT (G) 64.98 × 56.5 · NVMe Base Duo kit (12 mm standoffs) · Duo assembly · Pi 5 power button by the USB-C · Pi 5 mechanical drawing